

Can We See Eye to Eye? A Call for Prioritising Eye Visibility in Co-Located Social AR/MR Collaboration Across HMD Architectures

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ABSTRACT

Eye contact and gaze are foundational to face-to-face collaboration, supporting joint attention and coordination. In co-located Augmented (AR) and Mixed Reality (MR), these mechanisms are disrupted by Head-Mounted Displays (HMDs), which can occlude the eyes. While XR research explored gaze visualisation as an explicit awareness cue, collaborators' eye visibility and to how HMD architectures shape gaze-based interaction remains underexplored.

This position paper frames eye visibility as a socially consequential dimension of co-located AR/MR collaboration and argues that the distinction between Optical See-Through (OST) and Video See-Through (VST) HMDs is central. Drawing on social psychology, conversation analysis, and XR research, we show that studies examine OST and VST in isolation, obscuring how degrees of eye visibility affect collaboration and user experience. We call for unified comparisons across face-to-face interaction, OST, and VST, and outline the conceptual, methodological, and technological advances needed for cross-architecture research on eye visibility in co-located Social AR/MR.

Index Terms: Augmented reality, mixed reality, gaze, eye contact, head-mounted display, optical see-through, video see-through

1 INTRODUCTION

Human communication is fundamentally multimodal – in face-to-face interaction, speech unfolds alongside a largely unconscious stream of non-verbal cues, amongst which eye contact and gaze between interlocutors are regarded as some of the most informative signals [22, 24, 3, 2, 5]. Gaze behaviour in social interaction is far from incidental – the direction, duration, and reciprocity of gaze regulate conversational turn-taking and convey attentional focus [11, 17]. Gaze also plays a pivotal role in collaboration by supporting joint attention, grounding, and real-time coordination [8].

In the immersive domain, co-located collaborative Social Augmented (AR) or Mixed Reality (MR) environments aim to enhance face-to-face collaboration, allowing multiple users to share and manipulate virtual content in the same physical environment [31]. Increasingly, these systems rely on Head-Mounted Displays (HMDs), offering more immersion than handheld devices by freeing users' hands and registering virtual information spatially [39].

At the same time, using HMDs in co-located settings creates a paradox. While immersive software is designed to enhance face-to-face collaboration, the worn hardware creates a physical barrier, obstructing an essential non-verbal cue – eye contact [13, 39, 7, 26, 10, 30]. Although an HMD may occlude the eyes and the periorbital region, this disruption is not uniform across devices. The two dominant HMD architectures, Optical See-Through (OST) and Video See-Through (VST), differ to which extent they preserve

or diminish physical wearer's eye visibility. OST HMDs typically preserve eye visibility through semi-transparent optics, whereas VST HMDs fully occlude the eyes, replacing the direct visual perception of the environment with camera-mediated passthrough.

Despite this clear architectural distinction, its implications for non-verbal collaboration in co-located AR/MR remain underexplored. Existing research has devoted substantial effort to visualising gaze as an explicit cue [39, 18, 21]. By contrast, the degree to which collaborators' eyes remain physically visible, and how this varies across HMD architectures, has received comparatively little focused attention. Where eye visibility has been examined, studies tend to consider isolated configurations, such as OST versus a non-HMD baseline or VST systems with specific mitigation techniques, rather than providing a unified account across architectures. We thus argue for the necessity for cross-architecture comparison, i.e., observing how the collaborative experience is shaped by the HMD architectures worn by collaborators, since key social dimensions of collaboration quality and user experience may vary depending on how collaborators can see one another, particularly with respect to eye visibility. For simplicity and methodological consistency with the reviewed literature, the perspective of an individual observing their collaborator is adopted throughout the paper.

In order to contextualise our argument that eye visibility should be taken into account as a research variable in co-located AR/MR collaboration, we synthesise insights from social psychology, conversation analysis, Human-Computer Interaction (HCI), and Extended Reality (XR) research. We first review established findings on eye contact and gaze in both non-immersive and immersive communication and collaboration. We then compare HMD architectures and their social implications, including approaches aimed at mitigating periorbital occlusion. Next, we survey existing research on eye visibility and gaze across different HMD architectures.

Building on this analysis, we argue that the lack of unified comparison across face-to-face, OST, and VST collaboration reflects deeper conceptual, technological and methodological challenges rather than an absence of relevance. We examine why cross-architecture comparison matters for understanding gaze-based communication, articulate a set of architecture-spanning assumptions that motivate such comparison, and discuss the practical constraints that have so far limited its feasibility. Finally, we outline what should change in order to enable future research on eye visibility across HMD architectures in co-located Social AR/MR.

2 GAZE AND EYE VISIBILITY IN FACE-TO-FACE EXCHANGE

Foundational work on eye contact and gaze as a component of social interaction establishes its role in managing feedback, affiliation, intimacy, and interpersonal engagement [3]. Subsequent research demonstrated that gaze plays a crucial role in conversation management, with speakers and listeners exhibiting different gaze patterns [22]. In face-to-face interaction, listeners tend to maintain relatively steady gaze towards the speaker, whereas speakers alternate between looking at the listener and averting their gaze. These asymmetries reflect interactional role differences and support the coordination of conversational flow, including the management of turn-taking, with direct gaze often occurring at moments of turn

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completion [11, 23]. Furthermore, the speaker's gaze can guide listeners' interpretation, allowing them to resolve temporarily ambiguities before receiving disambiguating spoken information [15].

Beyond conversational regulation, gaze is a primary mechanism for establishing joint attention, i.e., the coordination of two individuals' attention towards a shared object, location, or event [1]. Studies employing simultaneous eye tracking show that the eye movements of collaborating partners become temporally aligned as they discuss and manipulate objects in a shared physical space [1]. Research on collaborative visual tasks demonstrates that sharing gaze, in this case visualising remote collaborator's gaze via a cursor on a monitor, can support efficient coordination by making attentional focus directly observable [8]. Accordingly, gaze coordination is not merely a by-product of collaboration, but an active mechanism for disambiguation and achieving common ground [15]. In a similar manner, research comparing head and gaze orientation shows that, while head orientation supports approximate spatial guidance, gaze visibility can provide finer-grained information that allows listeners to resolve referential ambiguity earlier and with less effort [16]. However, gaze also introduces additional processing costs, as listeners must divide attention between the speaker's face and the shared workspace, making it a more precise, but non-cost-free cue.

The social and perceptual consequences of occluding the periorbital region are also documented. The periorbital region here implies eyes and the immediately surrounding upper facial area [14], which is also the region typically covered by HMDs when worn. Research comparing eyeglasses and sunglasses shows that, while clear lenses have minimal impact on social perception, sunglasses, which obscure the immediate eye region through tinted lenses, hinder face recognition and reduce perceived trustworthiness [12]. Extending this line of work, a comparison of sunglasses with other facial occlusions demonstrates that occluding key facial regions, including the eyes, also hinders recognition and social trait judgments such as trustworthiness, competence, and attractiveness [4].

3 GAZE AND EYE VISIBILITY IN IMMERSIVE EXCHANGE

The importance of gaze has not been overlooked in XR research [36, 42, 32]. However, larger attention has been given to visualising gaze as an explicit virtual cue, in both remote and co-located collaboration scenarios [39, 18]. For example, virtual representations of the gaze can be used to show a remote collaborator's focus of attention, finding that such cues improve task performance and co-presence [35]. In co-located collaborative settings, bi-directional gaze visualisations, including gaze-ray, gaze-cursor, and gaze-trail techniques, facilitate joint attention, increase the use of deictic language, and are perceived as more engaging [21]. Similarly, augmenting collaborators with gaze visualisation, or other visualisations such as head direction, or view frustum, is preferred over the cue-less baseline and leads to improved collaborative performance [34]. While these approaches show the benefits of knowing where a collaborator is looking, they largely reduce gaze to a one- or two-dimensional pointer, overlooking the richer social information conveyed by seeing a collaborator's eyes in face-to-face interaction.

A complementary line of research indicates that HMDs can disrupt co-located collaboration by obscuring the periorbital region, limiting access to eye cues [26, 30, 37]. Furthermore, techniques for restoring periorbital cues occluded by HMDs have also been researched [26, 28, 27, 29, 7], further detailed in Sec. 6 and Sec. 5.

4 HMD ARCHITECTURE DISTINCTIONS – OST VS VST

The HMD architecture intrinsically determines the possibilities for non-verbal communication, regulating whether eye cues and eye-region information is directly available, even if degraded, or must be reconstructed through technological mediation. As the two dominant architectural paradigms, OST and VST present fundamentally different trade-offs between preserving direct visual perception of

the physical world and enabling full virtual augmentation, with important consequences for gaze-based social interaction [20, 30].

OST HMDs employ transparent optical combiners, such as waveguides or partially reflective mirrors, overlaying digital content onto the user's direct view of the real environment [20, 43]. Virtual elements are rendered as semi-transparent overlays that optically coexist with the physical scene rather than fully replacing it. Representative devices include the Microsoft HoloLens series, the Magic Leap series, XREAL Air and One series, and Snap Spectacles, which differ substantially in optical design, field of view, display brightness, and the degree of periorbital occlusion [20, 9, 33]. However, the promised transparency is inherently constrained by current optical requirements. To ensure sufficient contrast for virtual imagery, many OST displays are tinted or incorporate reflective coatings, which reduce eye visibility under common lighting conditions, as well as frames and sensor housings often partially occluding the periorbital region [20, 9]. As a result, interacting with an OST HMD wearer could be described as socially analogous to someone wearing tinted sunglasses, where eye contact remains possible but is diminished and ambiguous [33]. From a social perspective, the primary advantage of OST HMDs is that a co-located collaborator can, at least in principle, see the wearer's eyes, preserving a degraded, but still physically direct eye access [6].

VST HMDs most often use built-in front-facing cameras to capture the surrounding environment, which is then displayed to the user on opaque internal screens after digital augmentation [38, 20]. In this architecture, the user no longer has a direct view of the physical world through the HMD, but instead, all visual perception is mediated through a camera-based representation, commonly referred to as passthrough. Commercially available devices following this paradigm include the Meta Quest series and the Apple Vision Pro. From the observer's perspective, VST devices appear as solid masks that fully occlude the eyes and the periorbital region, thereby severing natural eye contact [13, 10]. This opacity removes direct access to the eyes and the periorbital region, as collaborators may be uncertain about the wearer's attention or engagement. Unlike OST systems, VST devices therefore provide no direct access to eye cues and necessitate explicit technological interventions to restore them. Nevertheless, even when gaze information is reconstructed, VST architectures introduce further challenges. The physical displacement of cameras relative to the user's eyes leads to geometric mismatches, while capture, processing, and rendering pipelines introduce latency [36]. As a consequence, camera-based representations of gaze cannot perfectly replicate the spatial precision or temporal dynamics of natural eye movements, complicating gaze interpretation and potentially diminishing social plausibility.

5 RESTORING EYE AND PERIORBITAL OCCLUSION

As previously discussed, while OST HMDs degrade eye visibility in varying degrees, VST HMDs typically remove that visibility entirely. This has motivated research that attempts to break the visual wall by reconstructing cues in the periorbital region through external displays, avatar substitution, or active HMD removal.

A key hardware-based approach involves external displays that project a digital representation of the wearer's eyes onto the front of the HMD. Apple's EyeSight feature exemplifies this by rendering a gaze-responsive image of the user's eyes on the exterior of the HMD [40]. Related research employs an external smartphone acting as a display on top of the HMDs [28, 27]. Later research has argued that reprojection onto an external 2D display suffers from incorrect depth cues and instead advocated for external autostereoscopic displays [29]. Furthermore, displaying stylised, cartoon-like eyes driven by real-time eye tracking is shown to improve a collaborator's sense of connectedness and ease of interaction [7].

One software-based approach overlays the real-life visual representation of the HMD in the observer's view with an avatar. Such

substitution can increase perceived mutual understanding and improve aspects of task performance [10], but it also highlights a trade-off between realism and robustness, as realistic avatars provide richer social cues, while being more susceptible to uncanniness than stylised options. Another software-based approach focuses on visually diminishing or removing the real-life visual representation of the HMD itself, for example, by masking the HMD and reconstructing the face region behind it [26]. However, such approaches remain constrained by sensing fidelity and alignment accuracy, and have largely been explored in controlled experimental settings.

All these strategies reveal a well-known challenge in immersive contexts – the uncanny valley [41, 26, 10]. Across external displays, avatar overlay, and HMD-removal approaches, issues such as imperfections and inconsistencies in facial rendering, processing latency, and virtual element tracking can result in unsettling or distracting outcomes, increasing cognitive load and potentially undermining the social benefits these techniques aim to restore.

6 COMPARATIVE ARCHITECTURE-DEPENDENT STUDIES

Despite eye contact's central role in face-to-face interaction, relatively few studies have directly examined how wearing HMDs affects eye contact and social interaction in co-located AR/MR collaboration. One of the earliest studies was done by Prytz et al. [37], examining collaboration of emergency responders with and without OST HMDs. Results showed that wearing an OST HMD reduced mutual gaze relative to a face-to-face baseline, but this reduction did not negatively affect subjective assessments of collaboration quality or task effectiveness. Although eye contact was significantly higher without the HMD, confidence and trust in the task artefacts were rated higher with the HMD, suggesting that the communicative cost of reduced eye contact is context-dependent, in this case involving highly trained professionals engaged in time-critical, role-defined tasks. Thus, it remains unclear whether similar reductions in eye visibility would be acceptable in more socially nuanced contexts.

More recent work has examined eye visibility from the perspective of social acceptability, rather than task performance. A study on OST HMDs with varying levels of lens opacity found that users preferred higher eye visibility, associating reduced visibility with unease, detachment, diminished social connection, and lower perceived trustworthiness [33]. Although this task did not involve immersive interaction, but rather participant pairs wearing OST HMDs and conversing, it indicates that even diminished eye visibility can negatively affect the social experience, aligning with earlier non-XR research [12]. In the scenario in which only a single collaborator is wearing an OST HMD, that information asymmetry is shown to create a power imbalance between the wearer and their collaborator, leading to a negotiation of social behaviours to compensate for the lack of transparency and agency [9]. Together, these studies indicate that even diminished eye visibility through OST devices can negatively affect the social experience, aligning with earlier non-XR research on tinted glasses [12].

In parallel, VST-based work by Kullmann et al. examines avatar-mediated restoration by compositing a personalised facial avatar either as a periorbital cutout within the HMD region or as full head substitution, compared against a non-augmented baseline with the same HMD [26]. Across two collaborative tasks, both techniques tended to speed consensus and increase perceived mutual understanding and behavioural interdependence, indicating that even imperfect avatar overlays can partially recover the communicative utility of gaze. However, the avatars were frequently rated as uncanny, and performance in the more demanding executive task deteriorated. Similarly, Vergari et al. compare co-located conversation in three conditions – unmediated face-to-face interaction, a conventional VST HMD (Meta Quest 3), and a VST HMD with an external display rendering the wearer's eyes (Apple Vision Pro with EyeSight) [40]. Interactions with EyeSight were rated as more so-

cially acceptable and engaging than with the non-augmented VST headset, although the face-to-face baseline still yielded the highest levels of social presence. While participants generally viewed EyeSight positively, some reported discomfort with the unnatural appearance and unsettling behaviour of the digital eyes.

In pursuit of our research objectives, Klinkert directly compared OST (Microsoft HoloLens 2) and VST (Meta Quest Pro) HMDs in a motor rehabilitation setting where only the patient wore the device [25]. While non-wearing therapists preferred the OST HMD for enabling better eye contact to assess the patient, their ability to communicate and judge the patient's condition was ultimately comparable for both devices, as they relied on other cues. The HMD-wearing patients themselves perceived no difference in their ability to make eye contact with the therapist, regardless of the HMD type.

After examining existing comparative studies, there remains a notable gap in the literature. Large focus is placed on isolated architectural comparisons, rather than examining **how different HMD architectures shape collaboration under comparable conditions**. As a result, there is an **absence of unified comparisons across face-to-face interaction, OST and VST collaboration**.

7 WHY CROSS-ARCHITECTURE COMPARISON MATTERS

Despite the existing investigations on the role of gaze and eye visibility in face-to-face collaboration, XR-centred research has largely examined OST and VST systems in isolation. The absence of unified comparisons across baseline, OST, and VST collaboration is therefore not only a methodological gap, but a conceptual one. Studying each architecture separately makes it difficult to unravel the effects of diminished eye visibility, degrees of eye occlusion, and technological mediation, as well as the adaptive interaction strategies that collaborators develop in response to each setup. As a result, architecture-specific studies risk normalising the constraints of the observed HMD architecture, producing findings that are internally coherent but difficult to compare or generalise.

A cross-architecture perspective is necessary because it enables a set of assumptions about non-verbal communication and collaboration that cannot be evaluated through isolated evaluations. In particular, comparing conditions of baseline interaction, OST collaboration, and VST collaboration within a unified frame could allow researchers to examine whether changes in behaviour and experience emerge gradually as eye visibility degrades, or whether shifts occur once direct visual perceptual access is obscured in its entirety. Our reasoning is guided by the following assumptions:

1. **Attention shifts from eyes to the shared workspace.** As access to periorbital cues diminishes across conditions, collaborators may look at one another less, since less actionable information is being conveyed. Visual attention is therefore more likely to be directed toward the shared workspace.
2. **Compensation through new social coordination strategies.** With attention shifted from the periorbital area, collaborators are expected to rely more on alternative coordination mechanisms, including non-verbal signals such as head orientation, and increased use of explicit verbal strategies such as naming and clarifications, particularly in VST conditions.
3. **Effects depend on task structure and role relations.** Differences between collaboration conditions are expected to be more pronounced in open-ended symmetric collaboration than in tightly structured, role-defined or time-critical tasks, where reduced eye contact may be more easily tolerated.
4. **Perceived naturalness, co-presence and social presence decline before performance.** Even when task performance remains stable, perceived naturalness, engagement, co-presence and social presence may decrease as eye visibility is diminished, helping explain why collaboration and user experience diverges from performance-based measures.

5. **Trust and attentional attribution questioned despite task success.** Reduced access to periorbital cues may increase uncertainty about a collaborator's attention and engagement in different collaboration conditions, shaping trust judgments even when collaboration succeeds.

Crucially, these assumptions cannot be meaningfully evaluated through isolated comparisons – only unifying baseline, OST, and VST conditions allows eye contact behaviour, collaboration strategies, and user experience to be interpreted as part of a continuum.

While this paper does not propose a single definitive experimental protocol, the following example might illustrate how cross-architecture comparison could be applied in practice. One could imagine a counterbalanced within-subject study comparing co-located interaction across three conditions: face-to-face, OST-mediated, and VST-mediated. The collaborative task could be based on the communication-focused survival game [19], adapted for this context by introducing shared virtual elements, as well as coordination points requiring mutual readiness. By leaving how agreement is signalled open, the task makes monitoring a partner's attentional state valuable for coordination, rendering eye contact a useful cue alongside other communication signals. While HMD-based measures such as partner-directed orientation and fixation duration can capture attempts to access eye cues at these moments, they cannot fully account for how (the lack of) eye contact is interpreted, particularly in the face-to-face condition. Complementary third-party observation, whether in real time or via video analysis, would therefore be needed to identify when and how collaborators engage in eye-based coordination, as well as the compensatory strategies that emerge. Brief post-condition self-reports could then contextualise these behaviours by assessing perceived attentional attribution, naturalness, and social presence across conditions.

8 FEASIBILITY OF CROSS-ARCHITECTURE RESEARCH

While the preceding discussion motivates the need to directly compare baseline, OST, and VST collaboration, the absence of such studies to date is not incidental. Rather, it reflects interconnected conceptual, methodological, and technical constraints that make cross-architecture evaluation difficult under present conditions.

OST and VST systems differ not only in eye visibility, but also in aspects such as field of view, motion-to-photon latency, visual fidelity, device bulk, and interaction techniques. Each of these factors can independently influence coordination, attention, and task performance. In a cross-architecture study, isolating the contribution of eye visibility from these co-varying properties is therefore challenging without either heavily constraining device capabilities or sacrificing ecological validity, which limits the feasibility of cleanly attributing observed effects to eye visibility and gaze alone.

Measurement asymmetry further complicates cross-architecture comparison. A limited number of VST HMDs include integrated eye tracking, enabling finer gaze estimation, whereas OST HMDs typically lack such abilities. As a result, one can either rely on different operationalisations of gaze across conditions or reduce all conditions to approximative proxies, such as head orientation. The former undermines comparability, while the latter goes against the theoretically meaningful distinctions between gaze- and head-directed attention, weakening the very constructs being studied.

Finally, it is important to reiterate that the social meaning of gaze is not constant across architectures. In baseline face-to-face interaction, eye contact norms are well established as part of everyday social behaviour. In XR interaction more broadly, the presence of technological mediation and augmentation can already alter expectations about how gaze should function, potentially weakening or reshaping norms carried over from non-immersive interaction. Within this space, OST interaction occupies an intermediate position – gaze remains visible, but in a degraded and often ambiguous form. In VST interaction, by contrast, gaze is augmented, simu-

lated, or occluded altogether. As a result, participants are not simply responding to differences in the amount of gaze available, but to different *expectations* about how gaze should function, complicating both experimental design and result interpretation.

9 RECOMMENDATIONS AND CHALLENGES

If cross-architecture investigation is to become tractable, several shifts are required. Methodologically, collaborative tasks should be designed to foreground the role of looking at collaborators, such as monitoring collaborator's attention, coordinating action, or establishing trust, while remaining interpretable when eye cues are diminished or occluded. This is particularly important for examining how collaborators shift from implicit, gaze-based coordination to more explicit strategies as eye visibility decreases, which are adaptations that are unlikely to emerge in tightly scripted tasks.

Technologically, greater control over eye visibility independent of device architecture is needed to mitigate the measurement asymmetries that currently limit cross-architecture comparison. For example, this may involve wider adoption of adjustable occlusion in OST devices, or research prototypes that decouple access to the eye region from specific commercial hardware, allowing eye visibility to be manipulated while keeping other factors constant.

Generally, baseline face-to-face interaction should be treated as an analytical reference point, while OST and VST should be understood as distinct conditions, rather than system variants. Framing eye visibility as a socially influential dimension, rather than a secondary implementation detail, creates space for comparative, architecture-aware co-located collaborative AR/MR research, in order to understand why and how collaborators look at each other.

Finally, another challenge lies in exploring heterogeneous collaborations. Instead of all collaborators wearing the same device, this includes not only setups with each participant wearing different HMD architectures, but also variations of devices within the same architecture. Such configurations introduce additional layers of technological, but also collaborative asymmetry.

10 CONCLUSION

In this position paper, we argue that eye visibility and gaze are socially consequential dimensions of co-located Social AR/MR collaboration that vary with HMD architecture. The identified lack of studies reporting unified comparisons across face-to-face, OST, and VST conditions reflects not an absence of relevance, but the conceptual, methodological, and technical challenges of comparing how collaborators can see each other and what implications that might impose, based on existing literature in the domain of social psychology and XR. We highlight how architecture-specific studies risk normalising device constraints and generalising differences in gaze behaviour, coordination, trust, and perceived naturalness. Rather than proposing a single definitive experiment, this position paper argues for HMD-based cross-architecture comparison as a necessary research direction, listing assumptions that motivate such empirical comparison. Clearer conceptual models of gaze, architecture-aware methodologies and experimental hardware that allow eye visibility are necessitated. Advancing this agenda is essential for understanding how immersive technologies reshape co-located collaboration.

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